STOCHASTIC INVENTORY OPTIMIZATION

A mini project on making stock decisions when demand is uncertain

Course: Stochastic Decision Making

Submitted by

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Note: The prices, costs, demand values, and probabilities in this report are assumed for an academic simulation. They are not taken from a real business.

Abstract

A shop has to decide how much stock to buy before it knows how many customers will arrive. If it buys too little, some sales may be missed. If it buys too much, some items may remain unsold. In this project, I model this decision for one product and compare different stock quantities using expected profit. Demand is represented by three possible scenarios, each with an assigned probability. I calculate profit for each scenario, then use Monte Carlo simulation with 100,000 trials to check the result and look at profit variation. Under the assumptions used here, stocking 40 units gives the highest expected profit among the quantities tested.

1. Problem and objectives

The main question is: **How many units should the shop stock to get the highest expected profit when demand is uncertain?**

The project has five practical objectives:

- Define a simple inventory decision and its profit calculation.
- Compare possible stock quantities across the demand scenarios.
- Find the quantity with the highest expected profit.
- Use Monte Carlo simulation to estimate average profit and its variation.
- Check whether the decision changes when the demand probabilities change.

2. Model assumptions

To keep the model manageable, I assume one product and one selling period. The selling price, purchase cost, and salvage value stay fixed. Any unsold unit can be recovered at the salvage value. The demand values and their probabilities are illustrative, rather than estimates from actual sales data. Stock quantities are tested from 0 to 80 units in steps of 5.

3. Data and mathematical formulation

The model uses the following assumed demand distribution:

Demand (units)	Probability
20	0.20
40	0.50
60	0.30
Total	1.00

The expected demand is 20(0.20) + 40(0.50) + 60(0.30) = 42 units. This is an average over the probability distribution; actual demand in a single period can still be 20, 40, or 60 units.

3.1 Decision variable

Let **Q** be the number of units purchased before demand is known. The model tests Q = 0, 5, 10, …, 80.

3.2 Profit calculation

Selling price = ■100 per unit; purchase cost = ■60 per unit; salvage value = ■10 per unsold unit. For stock quantity Q and demand D, profit is:

Profit(Q, D) = 100 × min(Q, D) + 10 × max(Q – D, 0) – 60 × Q

The first part is revenue from items sold. The second part is money recovered from leftover stock. The final part is the cost of buying all Q units. For example, if Q = 40 and D = 20, revenue is ■2,000, leftover value is ■200, and purchase cost is ■2,400. Profit is therefore –■200.

3.3 Objective

For each candidate Q, expected profit is calculated by multiplying the profit in each demand scenario by its probability and adding the results. The chosen quantity is the candidate with the largest expected profit.

4. Method followed

I used a scenario calculation first, because it is easy to check and gives an exact expected-profit value for the assumed distribution. Monte Carlo simulation was then used as a second check.

Step 1 — Set the candidate quantities

The code checks quantities from 0 to 80 in increments of five units. This creates a manageable set of possible decisions.

Step 2 — Calculate profit in each scenario

For each quantity, the profit function is evaluated at demand values 20, 40, and 60.

Step 3 — Calculate expected profit

Each scenario profit is multiplied by its probability. The three weighted values are added to get expected profit for that stock quantity. The quantity with the largest value is selected.

Step 4 — Run the Monte Carlo simulation

The simulation randomly draws demand values using the stated probabilities and calculates profit for the chosen quantity. It repeats this process 100,000 times. A fixed random seed is used so the results can be reproduced when the same code and library behaviour are used.

Step 5 — Check sensitivity

Finally, I vary the probability assigned to high demand and recalculate the preferred quantity. This gives a simple view of whether the decision is sensitive to the assumed demand pattern.

Tools used

Python is used for the calculations. NumPy supports random sampling and numerical operations, pandas stores results in tables and CSV files, and Matplotlib creates the charts. The repository contains the Python script, Jupyter notebook, generated results, and this report.

5. Results

The highest expected profit in the tested range occurs at a stock quantity of **40 units**. With the assumed selling price, cost, salvage value, and demand probabilities, its exact expected profit is ■1,240 per period. The Monte Carlo estimate is close to this value, at approximately ■1,241.42. Small differences are expected because random simulation estimates an average from sampled outcomes.

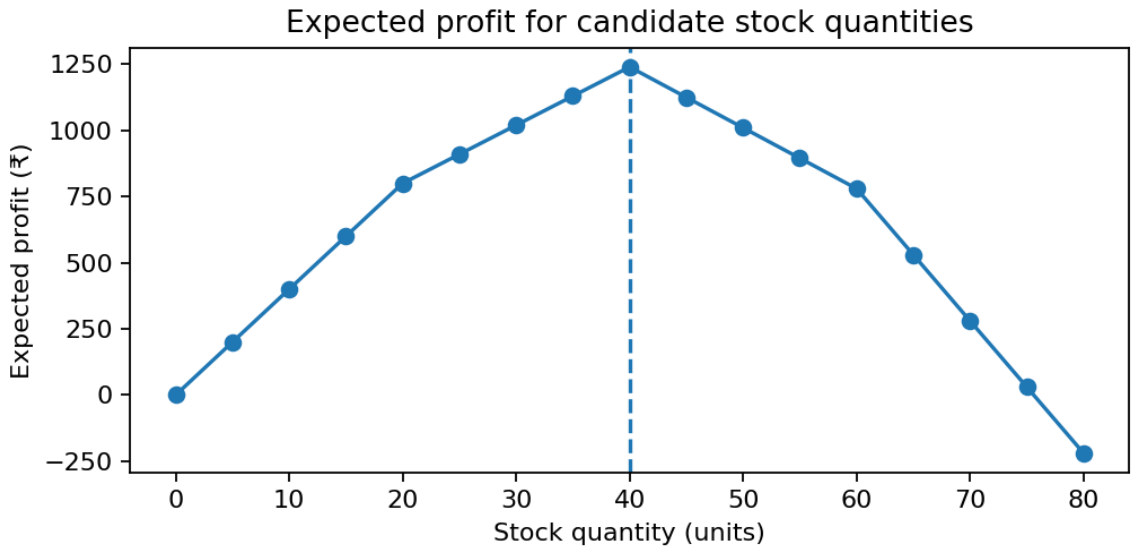


Figure 1. Expected profit across the candidate stock quantities.

The curve increases as stock is raised from low quantities, reaches its maximum at 40 units, and then decreases. When stock is above the most useful level, the extra purchase cost is not fully recovered by sales or the salvage value.

At Q = 40, the scenario profits are:

Demand	Probability	Profit at Q = 40
20 units	0.20	–■200
40 units	0.50	■1,600
60 units	0.30	■1,600

Expected profit = (0.20 × –200) + (0.50 × 1,600) + (0.30 × 1,600) = ■1,240.

6. Simulation and sensitivity analysis

The simulation repeats the demand draw many times and records the profit for each trial. Along with the mean, the code records measures such as standard deviation and the 5th and 95th percentiles. These values describe the spread of possible outcomes under the model; they are not guarantees about a real shop.

6.1 Sensitivity check

I changed the probability of high demand and recalculated expected profit for each candidate quantity. In the tested settings, 40 units remained the best quantity. The maximum expected profit increased as more probability was assigned to high demand, because the higher-demand scenario became more likely. This result depends on the other assumptions remaining fixed.

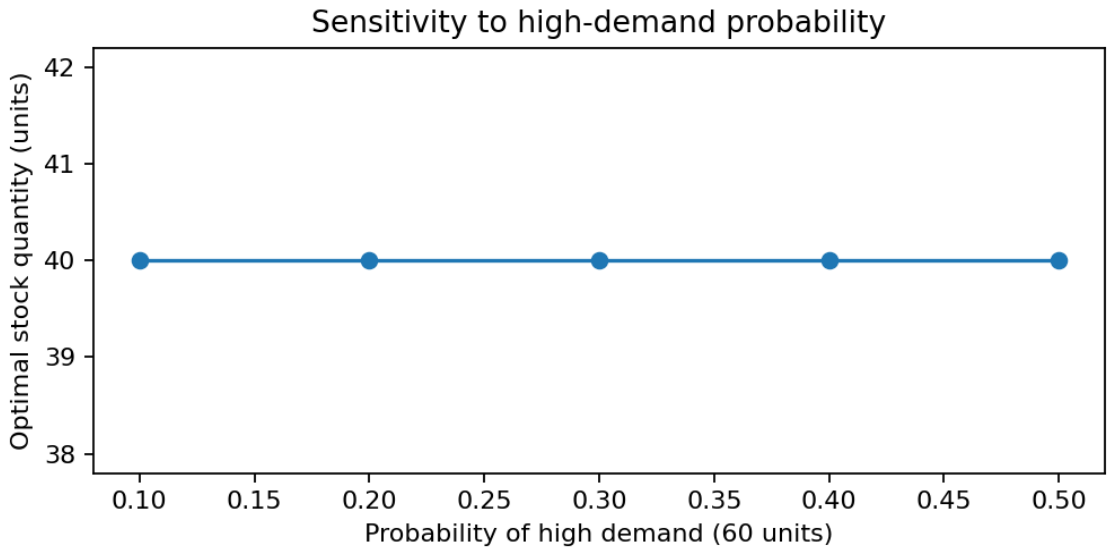


Figure 2. Optimal quantity under changes to the high-demand probability.

This is a small sensitivity exercise, not a complete study of every possible demand distribution. A real analysis should estimate probabilities from historical sales and test a wider range of costs, prices, and demand patterns.

6.2 Why use both calculations?

The scenario method gives an exact expected value for the three assumed demand outcomes. Monte Carlo simulation is useful when a model becomes more complicated or when there are many uncertain inputs. Comparing the two here is a simple way to check that the simulation output is reasonable.

7. Limitations and conclusion

7.1 Limitations

• Only three demand values are considered, and their probabilities are assumed.
• Price, purchase cost, and salvage value are fixed.
• The model covers one product and one selling period.
• It does not include storage limits, supplier lead times, replenishment, spoilage, or changing prices.
• Stock quantities are restricted to multiples of five between 0 and 80.

For these reasons, the result should be treated as an example of the method rather than a recommendation for an actual business.

7.2 Conclusion

This project helped me compare inventory decisions using a simple probability model instead of relying only on average demand. I evaluated candidate quantities, calculated expected profit, and used Monte Carlo simulation to check the result and describe profit variation. For the assumptions in this project, 40 units gives the highest expected profit of ■1,240 per period, with a simulation estimate of about ■1,241.42. The sensitivity check also showed that the best decision depends on the demand distribution and the costs used. If this model were applied to a real retailer, the first improvement would be to replace the illustrative assumptions with actual sales and cost data.

References

1. Nahmias, S. and Olsen, T. L., *Production and Operations Analysis*, topics on inventory management and stochastic models.
2. NumPy documentation: https://numpy.org/doc/
3. pandas documentation: https://pandas.pydata.org/docs/
4. Matplotlib documentation: https://matplotlib.org/stable/
5. Course assignment handout, Mini Project — Stochastic Optimization Model, Stochastic Decision Making.

Reproducibility note: Results in this report come from the project code and the illustrative assumptions listed above.